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HANDBOOK OF SURVEY RESEARCH

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Chapter 10

The Psychology of Survey Response

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Over the last 25 years, many survey methodologists have adopted the assumption that reporting errors in surveys largely reflect problems in the cognitive processes through which respondents generate answers to survey questions. For example, respondents may misunderstand the questions or have trouble remembering the information needed to formulate an accurate answer. These cognitive processes are often grouped under four major headings — comprehension, retrieval, judgment and estimation, and reporting — and each of these sets of processes has its characteristic pitfalls. This chapter describes the cognitive approach to reporting error, outlining what we know about how respondents answer survey questions and how those processes can go awry. It also briefly characterizes alternative approaches to the understanding of reporting errors, such as conversational analysis.

10.1. Introduction: Models of the Survey Response Process

Although surveys have had a prominent place in American life at least since the 1930s, models of the process by which respondents come up with answers to the questions in surveys are much more recent. The earliest discussions of the response process focused on the social nature of the survey interview and presupposed an interviewer-administered questionnaire. (In the last couple of decades, various forms of computerized self-administration have become popular, but many surveys, especially those done by academic and government researchers, still involve an interviewer and respondent and their interacting to get the survey questions answered.) For example, in their classic work, *Response Effects in Surveys*, Sudman and Bradburn (1974, p. 6) argued that the survey interview constituted “a microsocial system in which there are two roles, that of respondent and that

of interviewer, joined by the common task of giving and obtaining information.” Their analysis of response errors was based on this social model of the interview and traced errors to the role behavior of the two actors. But the social model, at least as articulated by Sudman and Bradburn, anticipated the cognitive approach by identifying the demands of the task itself as the main source of error.

In addition, the social model of the interview situation and its impact on survey errors anticipated other recent developments in the survey methods literature, notably the careful and detailed analyses of the interaction between respondents and interviewer that apply conceptual tools from the literature on conversational analyses to survey interactions (e.g., Maynard, Houtkoop-Steenstra, Schaeffer, & van der Zouwen, 2002; Schaeffer, Dykema, & Maynard, this volume). The social model outlined by Sudman and Bradburn also anticipated later cognitive and conversational models of the survey response process in emphasizing how norms from other everyday settings can affect the interview. Sudman and Bradburn emphasized such norms as telling the truth and saving face. People have a lifetime of experience in answering — and fending off — questions, and the norms and expectations distilled from this wealth of experience affect respondents’ behavior in the interview setting.

The cognitive approach to reporting errors in surveys dates back to the early 1980s. During that period, survey methodologists had recognized that survey errors arose in part because respondents omitted events they forgot completely and reported events they shouldn’t have because they misremembered when the events had taken place. As early as 1964, a classic paper by Neter and Waksberg (1964) presented a sophisticated four-process model of reporting error in a survey on spending on home repairs. The processes described by Neter and Waksberg — forgetting, errors in dating the repairs, failures to report incidents due to repeated interviews (“conditioning”), and omissions reflecting the burden of reporting — are still apparent in contemporary discussions of the cognitive underpinnings of reporting errors. Based on their growing awareness of the cognitive sources of reporting problems in surveys, survey researchers initiated a series of conferences in the 1970s and 1980s that brought together cognitive psychologists (especially those who were studying memory) and survey methodologists to explore how concepts and methods drawn from cognitive psychology could be applied in solving survey methods problems, particularly issues of reporting error. The third of these conferences (the Advanced Seminar on Cognitive Aspects of Survey Methodology, or the CASM Seminar) in 1983 led to the widespread acceptance of a heuristic model that argued that the survey response process included four main components — comprehension of the question, retrieval of relevant information, integration of this information via judgment or estimation processes, and reporting of the resulting judgment or estimate. This chapter follows this model (which we will refer to as the CASM model) in presenting the key results from the cognitive approach to survey error. We note that, although it is tempting to think of these four components as occurring in sequence, there are good reasons to think that the steps overlap in time (retrieval begins before comprehension is complete; information is integrated even as further information is retrieved; and so on). Further, respondents may sometimes

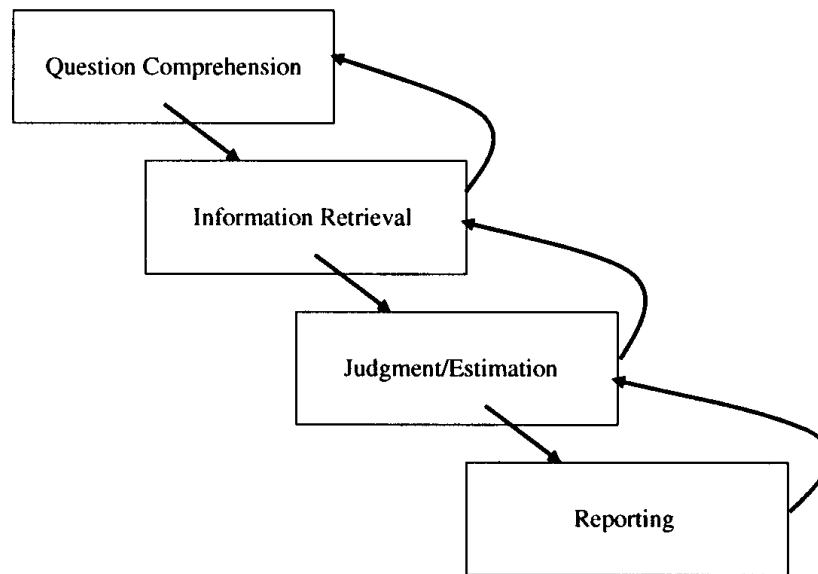


Figure 10.1: A simple model of the response process for survey questions.

backtrack from a “later” component to an “earlier” one. Figure 10.1 depicts the CASM model.

There are several alternatives to the four-component CASM model that still fall under the cognitive banner. The earliest of these is the model proposed by Cannell, Miller, and Oksenberg (1981); their model distinguished two main routes that respondents take to arrive at their answers. Some respondents take the more effortful path, following a sequence of cognitive steps: they first comprehend the question, next retrieve and organize relevant cognitions, then evaluate their potential answers for their accuracy, and finally report answers that are based on reasonably thoughtful cognitive processing. (This path clearly resembles the four-component model that came out of the CASM Seminar.) But some respondents follow a less effortful path, in which their answers are based on superficial cues available in the interview situation, such as the observable characteristics of the interviewer (e.g., his or her race) or the prior questions in the survey. Such answers are prone to a number of response problems, including social desirability bias (the tendency to present oneself in a favorable light) and acquiescence (the tendency to agree with an assertion, regardless of its content). Cannell’s work tended to emphasize the importance of respondent motivation and it explored various techniques to keep respondents from straying from the more effortful path.

Krosnick’s (1991, 1999) more recent satisficing model is a sophisticated descendant of the Cannell, Miller, and Oksenberg approach. His model distinguishes two modes of responding — optimizing and “satisficing.” (The notion of satisficing was originally proposed by Herbert Simon to describe how certain decisions are made.) Optimizing resembles the more effortful of the two paths distinguished by Cannell and his colleagues. Optimizers carefully attend to the meaning of the question, make a thorough search of their memories, integrate what their search process turns up using unbiased judgment or estimation strategies, and then honestly

report their answers. This careful processing contrasts with satisficing. Satisficing respondents either carry out these processes sloppily (weak satisficing) or skip the retrieval step entirely (strong satisficing). We take a closer look at Krosnick's satisficing model in Section 10.3.6.

One other model of the response process is worth mentioning. It was proposed by Strack and Martin (1987) to deal with answers to attitude questions. The model distinguishes between respondents who already have a judgment about the issue or person in question and those who have to construct a new one. Those with an existing judgment retrieve that judgment and report it, thus skipping the integration component of the response process. Those who don't already have a judgment form one after they retrieve various relevant considerations about the issue or person; the response process for this latter group includes all four components making up the CASM model.

For a variety of reasons, we follow the CASM model in our discussion of the response process. The various "two-track" models of the survey response process certainly make useful distinctions, but the alternative tracks they postulate are readily viewed as variations on the four-component theme and, as we will argue, for many types of survey items, two tracks are too few to capture the key differences in how respondents arrive at their answers.

10.2. Cognitive Processes in Answering Survey Questions

Although the various models of the survey response process differ in many details, they generally agree on the major processes respondents go through (or ought to go through) in answering questions. These processes are: (1) comprehending the meaning of the question; (2) retrieving relevant information; (3) formulating an answer; (4) formatting that answer to meet the requirements of the question and perhaps "editing" it to satisfy other motives the respondent may have. As we already noted, it is tempting to view these processes as unfolding in a linear sequence, but most researchers recognize that the processes occur in the flow of a conversation and they may go on in parallel or rapidly cycle back and forth (see Figure 10.1). For simplicity of exposition, we will present them as though they were distinct steps that proceeded in an orderly sequence.

10.2.1. Comprehension

To answer a question, respondents must first understand what they are being asked. For the researcher the goal is to get respondents to understand the question in the same way that the researcher does. In practice, this goal can be very difficult to achieve because of the many subtleties and ambiguities of language. Indeed Belson (1981), who did an extensive study of respondents' understanding of common terms

such as “weekday,” “children,” “regularly,” and “proportion,” found widespread misunderstanding of these and other common terms in survey questions.

Comprehension begins with respondents identifying a string of sounds or written symbols as words in a language they understand. In addition to these more or less perceptual processes, comprehension can be analyzed on at least three levels — as a syntactic, semantic, and pragmatic process. The syntactic level encompasses mapping the words and their grammatical relations onto some underlying representation that captures their logical relationships. The string of words that make up the question is “parsed” into syntactical units — that is, the words are grouped into meaningful units and meaning is extracted from these units by a process that is still poorly understood. A key component of this grammatical analysis of the question is the identification of the question’s focus and presupposition. Every question presents an incomplete picture of some state of affairs and asks the recipient of the question to fill in the missing piece. The partial picture offered by the question is its *presupposition* and the missing piece that the respondent is supposed to supply is the question’s *focus*. The syntactical analysis of the question can be difficult if the question is syntactically ambiguous. The sentence “Flying planes can be dangerous” is structurally ambiguous — it is compatible with two different underlying representations, depending on whether “flying” is understood as a verb or as an adjective.

In addition to conducting this structural reading of the question (and in parallel with it), the respondent must carry out various semantic processes, including retrieving the intended meanings of individual words. This can be complicated since the words may be relatively unfamiliar to the respondent. In addition, many, perhaps most, words have multiple meanings, a situation referred to as lexical ambiguity. For example, the question “Where is the table?” is lexically ambiguous because the word “table” can refer to an object on which things can be placed or a set of numbers arranged in labeled rows and columns. Lexicographers distinguish between homonyms — words like “bank” with several completely unrelated meanings (the natural incline at the side of a river and the financial institution) — and polysemous words — words like “paper” with multiple related “senses” (a paper is both a certain kind of academic prose and the material it is printed on). Such lexical ambiguities are inherent in language and are usually resolved by the context within which the sentences appear. In some cases, the words may not be ambiguous but respondents may vary in how they interpret them or map them onto their own specific situation. For example, a question that asks whether the respondent has purchased “furniture” may be unclear to a respondent who has recently purchased a floor lamp and isn’t sure whether a floor lamp counts as furniture. Almost every category, including furniture, includes relatively clear instances (a sofa or table) and borderline cases (floor lamps) that aren’t clearly members of the category but aren’t clearly nonmembers either. Respondents may be unsure how to classify these borderline instances.

Context plays an important role not only in resolving ambiguities but also clarifying the meaning of words that are unfamiliar. For example, a study by Tourangeau and Rasinski (1988) found that a question about the Monetary Control

Bill, an obscure piece of proposed legislation, was interpreted as referring to an anti-inflationary measure when it occurred after a series of questions about inflation; in fact, the bill concerned the regulation of the banking industry.

The pragmatic level of analysis involves respondents' understanding of the underlying intent of the question. Questions are often used for other purposes besides requesting information. For example, when someone asks "Does it seem cold in here?" this is probably better understood as an indirect request (to close the window or turn down the air conditioner) than as a question; answering "Yes, it does" would usually be seen as unresponsive. Listeners, including survey respondents, follow various strategies in ferreting out the intent of a question. Grice (1975) argued that the participants in a conversation follow four general maxims that collectively provide a basis for filling in many of the gaps between the words that are uttered and the intent behind them:

1. *The maxim of quantity*: Provide as much information as is necessary for the purpose at hand and no more;
2. *The maxim of quality*: Be truthful;
3. *The maxim of relation*: Stick to the topic;
4. *The maxim of manner*: Be clear.

When questions or other utterances seem to violate these maxims, respondents may reinterpret them. For example, Schwarz, Strack, and Mai (1991b) showed that when respondents are asked about their marital happiness and then asked a similar question about their overall happiness ("Taking things altogether how would you say things are these days? Would you say you are very happy, somewhat happy, or not too happy?"), they interpreted the latter item as though it were intended to *exclude* their marriages (see also Tourangeau, Rasinski, & Bradburn, 1991); otherwise, the pair of questions would have seemed too redundant, violating the maxim of quantity. Because the respondents assumed the questions followed Grice's maxims, they interpreted the questions in a way that minimized the overlap in their meaning.

10.2.2. *Retrieval*

Once a question has been comprehended, respondents must retrieve from memory whatever information is needed to formulate their answers. Cognitive psychologists distinguish between two major memory systems — long-term memory and working memory; both are involved in retrieval.

When lay people think about memory, they are mostly thinking about long-term memory, which is the system that stores autobiographical memories (memories for our experiences), along with other types of information. Long-term memory has a very large capacity. It appears to encompass two rather distinct subsystems, semantic memory and episodic (or autobiographical) memory. Semantic memory refers to memories associated with vocabulary, language structure, and general knowledge of

the world, whereas episodic (or autobiographical) memory refers to memories for specific events that took place in a particular time and place. Both of these systems consist of declarative knowledge that is (or can be) represented in a propositional format. Long-term memory also includes procedural knowledge, or our knowledge of how to perform various activities, like riding a bike, playing chess, or tying our shoelaces. Procedural knowledge (knowing how in contrast to knowing that) is likely to be stored in a different format from declarative knowledge.

Working memory refers to the system in which active cognitive processing takes place. Working memory includes information we are consciously aware of, as well as other information from long-term memory that is currently being used by some ongoing cognitive process. Many processes take place so quickly that we remain unaware of them and of the information they are acting on. For example, most of us are unaware of the processes involved in the comprehension of language; these processes mostly take place outside of awareness. (We are, of course, aware of the products of these processes.) Working memory has limited capacity but rapid access, and it can be overtaxed by demanding cognitive tasks. By contrast, long-term memory has a large, essentially unlimited capacity, but is relatively slow in access. The process of retrieval consists of moving information from long-term memory into working memory (or, to put it a different way, retrieval means raising the activation of material in long-term memory to a high enough level that it becomes conscious).

Before we discuss the retrieval process in more detail, it is useful to examine the structure of autobiographical memory, the portion of long-term memory that is most frequently relevant to survey questions. Autobiographical memories appear to have an organization that is at least in part hierarchical. For example, Barsalou (1988) argues that there are four levels in this hierarchy. At the top level are various important life domains, such as one's employment history or residential history. Within each life domain, there are various extended time periods (one's first job, one's jobs during college, etc.); these are, in effect, the major eras in one's life. At the next level are memories of individual events. Alongside these individual event memories are more generic memories that capture the common features of many similar events; Barsalou refers to these as "summarized events." Finally, at the lowest level, there are the details about the specific events, including perceptual information (such as visual images). This sensory information at the lowest level may be stored in a separate subsystem.

Retrieval, or remembering, is a process in which we search long-term memory for the information needed to answer a question. If we think of long-term memory as a big storehouse, it is clear that it must be organized in some way in order for us to be able to retrieve things from it. Just as we must label files when we put them in file drawers, so we must attach some kind of labels to information in long-term memory. The labeling process, often called "encoding," refers to various aspects of the information or experience, including its emotional tone, attached to the item when we store it in memory that allow us to retrieve it. (For a more complete discussion of memory models, see Tourangeau, Rips, & Rasinski, 2000, Chapter 3).

The process of retrieval has both automatic and controlled components. The automatic portion (which happens outside of our awareness and is beyond our

control) involves one memory or idea bringing to mind related ideas. If we think of long-term memory as a network of linked concepts, then activating one concept tends to activate closely linked concepts as well. This spreading activation process typically begins with one or more retrieval cues — the information given in the question and any explanatory material available to respondents at the time they are asked to recall an event. Retrieval cues are any words, images, emotions, etc. that activate or direct the memory search process. The controlled portion of the process involves deliberately generating retrieval cues that might bring to mind the sought-after memories. For example, the National Crime Victimization Survey asks respondents: “During the last 6 months, (other than any incidents already mentioned) did you call the police to report something that happened to YOU which you thought was a crime?” If the cues in the question fail to trigger the retrieval of a memory, a respondent might think about specific times or places where he or she might have reported a crime (say, at his or her workplace or while he or she was on vacation). The cues in the question and any that the respondent generates activate the pathways of association leading to the desired information.

10.2.3. Judgment and Estimation

Based on the information they retrieve, respondents must formulate an answer to the question. The exact processes involved differ from one question to the next, but across a range of survey questions, Tourangeau, Rips, and Rasinski (2000) have argued that these processes fall into four main categories:

1. Direct retrieval of an existing answer;
2. Retrieval of general information from which an answer can be constructed or inferred;
3. Retrieval of specific memories from which an answer can be derived; and
4. Retrieval of an impression.

For some questions, respondents already know the answer and if that answer is highly accessible — that is, easy to retrieve from long-term memory — it may leap to mind when the question is asked. When the questions ask about well-rehearsed topics, such as one’s date of birth or marital status or about attitude issues for which the respondents have an existing well-articulated position, the respondents may retrieve the answers directly. The answers spring fully formed from memory and can be reported directly.

In many cases, though, the questions don’t tap some previously formulated answer. Most of us probably keep a running tally of how many children we have or how many cars we own, but we are less likely to keep tabs on the number of visits we made to the doctor in the last six months or the number of times we’ve used an ATM in the last four weeks. When we are asked about such matters, one possibility is that we retrieve something generic (e.g., the fact we go to an ATM once a week) and use that generic information to make an estimate (cf. Burton & Blair, 1991, on rate-based

estimation). Similarly, in dealing with an attitude question (say, about opening up the Arctic National Wildlife Refuge to oil drilling), we may not have a ready-made judgment on the issue, but can cobble a position together from more general values (such as the importance of preserving the environment or of promoting energy independence). We can think of these strategies for getting to an answer as top-down — that is, they involve reasoning from something general (like a rate or a general value) to the specific case at hand.

Other times, we seem to reason from the bottom up. When they are asked about the number of times they have visited a doctor in the last six months, some respondents may recall each individual visit and count them up, a strategy sometimes referred to as recall-and-count (or episodic enumeration). Similarly, respondents may construct their judgment about a specific political candidate by recalling what they know about the candidate, evaluating each of those facts, and then integrating these evaluations into an overall judgment.

A final group of strategies is based on the retrieval of some relevant impression. For example, we may not know how often we use an ATM each week, but we may have some general sense (“I rarely go” or “I go all the time”). Similarly, evaluations of political candidates (or people we have just met) often seem to be based on vague impressions and it can be difficult to recall the specifics on which these impressions were based.

Which particular path respondents take in reaching an answer depends in large part on what they retrieve in response to the question. What gets retrieved in turn reflects what information in long-term memory is chronically accessible (chronically easy to retrieve) but also what information is temporarily accessible because it has been activated by the question itself, by prior questions, or by other contextual cues. In addition, there is some evidence (e.g., Burton & Blair, 1991) that respondents sometimes deliberately switch from relatively low-burden strategies (such as rate-based estimation) to more difficult ones (such as recall-and-count) when they think the question is important and they have enough time (see also Sanbonmatsu & Fazio, 1990, who make a similar argument regarding attitude judgments).

Because most answers to survey questions are constructed (rather than retrieved), they may be affected by information that is temporarily accessible. With attitude judgments, the prior questions may bring to mind specific considerations regarding an issue or some general value related to it, resulting in assimilation effects, in which the judgment is distorted in the direction of earlier judgments (Schwarz & Bless, 1992; Tourangeau & Rasinski, 1988). The opposite (contrast effects) can also happen. Prior items may serve as a standard of comparison to which the current one is compared. Or respondents may disregard information that is temporarily accessible because they want to avoid being redundant or because they see the information as irrelevant to the judgment at hand; when respondents throw out accessible information they might otherwise have taken into account, contrast is the result. The size of these assimilation and contrast effects depends on the amount and extremity of the temporarily accessible information that is used or discarded in arriving at a judgment.

10.2.4. *Reporting an Answer*

Surveys not only pose the questions but they often impose various requirements on the form of the answers. Thus, the respondents have to fit whatever judgment or estimate they reach into the response format that the questionnaire offers. Rarely do surveys allow respondents to answer questions in a free (or open) format. Open-ended questions have a multitude of problems, not least of which is the cost and difficulty of transforming free-form answers into a format that can be treated quantitatively. Today, almost all questionnaires depend on closed or precoded questions. And the open-ended questions that are included in surveys are still likely to require answers in a circumscribed format. For example, a question may ask for a numerical answer ("How many children live here?") and the interviewers will only accept numbers that take the right form and fall within a specified range (e.g., whole numbers between zero and 20).

Relatively little is known about how respondents format (or *map*) their judgments and estimates so that they meet the requirements of the questions. With rating scales, it seems likely that respondents define the end points of the scale in terms of the most extreme instances of the category to which the object of judgment belongs; that is, even when we are asked to rate a single object, the judgment is implicitly comparative (we judge one object relative to others of the same type) and the extreme instances that come to mind help anchor the rating scale. As we shall see, respondents have considerable difficulty mapping their judgments onto scales that are not clearly anchored. When the possible answers do not form a scale, respondents appear to evaluate the options in sequence, with the options they consider first getting more attention than those that come later. Which options respondents consider first depends on the mode of data collection. When the survey questions are administered visually (as in a web survey or self-administered paper questionnaire), respondents seem to start at the top of the list. When the questions are administered orally (as in a telephone survey), they seem to start with the last option they heard (Krosnick, 1999).

Respondents don't always report the answer that best fits their situation. There are several reasons why they "edit" their answers in this way, but the most common one is to avoid embarrassing themselves in front of an interviewer. As a result, respondents tend to report doing things that they haven't done but feel that they should have (such as voting or attending church) or to deny doing things that they have done but feel that they shouldn't have (such as illicit drug use). Tourangeau and Yan (2007) provide a comprehensive review of tactics that survey researchers have used to reduce such "social desirability" biases; one of the most common is reduce the role of the interviewer by having the respondent complete a self-administered questionnaire (either on paper or via computer).

10.3. Problems in Answering Survey Questions

The major payoff of the foregoing analysis of the response process is that it leads to a more systematic catalogue of the ways in which the process can go awry.

In this section, we examine eight categories of potential pitfalls in the response process, starting with a process — encoding — that occurs before any question is asked and ending with one — navigating a self-administered questionnaire — that occurs after the respondent answers one item and is attempting to figure out which question to answer next. Not surprisingly, the other categories of errors involve problems in comprehending questions, retrieving information from memory, making factual estimates or attitudinal judgments, getting the answer into the required format, and editing it.

10.3.1. Encoding Issues

Encoding refers to how we take in information in the first place, how we experience events, interpret them, and store them in memory. The encoding process can give rise to two major problems for surveys. First, respondents may not take in the information that the survey researchers want to ask about. Second, respondents may encode the relevant information but not in the terms that the researchers use.

A study on parents' recall of their children's immunizations illustrates the problems that arise when respondents never take in the requested information in the first place. Lee and her colleagues conducted a series of studies in which interviewers asked parents (or other caretakers, such as grandparents) as they were leaving the pediatrician's office about what immunizations their child had just received (Lee et al., 1999). Although memory burdens were at a minimum, Lee and her colleagues found in their first study that respondents were only at about chance levels of accuracy. For three of the immunizations (those for diphtheria, tetanus, and pertussis; for hepatitis B; and for *Hemophilus influenzae* type b disease), the reports had essentially no correlation with the records showing which immunizations the children had actually received. There are several reasons why respondents may have failed to take in the information about which shots their children got: the immunizations have long, unfamiliar, and complicated names; the child often receives multiple immunizations in a single visit; the diseases themselves are unfamiliar (thanks in part to the success of the immunizations); immunizations are a routine feature of well-baby care; the respondents probably count on the doctor or nurse to administer the right shots and attend to other matters (such as comforting the child) instead; and, finally, there is no particular reason for respondents to remember the information since it will be recorded on the immunization card that the respondent receives. (Lee and her colleagues asked respondents not to consult the immunization card as they answered.) All in all, the information about immunizations is difficult for most people to encode and they have little motivation to attend to it carefully; as a result, they don't encode much information about which shots their children received on any particular occasion.

The second problem that can be traced to the encoding process is that the information respondents do take in may not be encoded in the same terminology that the survey question uses. For example, respondents may not think of their doctors as

“primary care physicians.” Here is a question (taken from the National Health and Nutrition Examination Survey) about exercise that may not make contact with respondents’ memories because of the mismatch between the terms in the question and those used by respondents in encoding their experiences: “How often do you do light or moderate activities for at least 10 minutes that cause only light sweating or a slight to moderate increase in breathing or heart rate?” It seems unlikely that someone who, say, swims three times a week encodes it as a “light to moderate activity” that causes “only a slight to moderate increase in breathing.” In consequence, he or she might not think of swimming in response to the question (see Tulving & Thomson, 1973).

One of the reasons why survey researchers conduct focus groups is to avoid these problems with encoding. Focus groups (Krueger & Casey, 2000) are a useful method for exploring what members of the survey population know about a given topic (that is, what they have encoded) and what terminology they use in discussing it (how they have encoded it).

10.3.2. Misunderstanding the Questions

As we noted earlier, the comprehension of a survey question involves syntactic, semantic, and pragmatic processing. For our purposes, the key syntactic processes involve mapping the question onto an underlying logical representation, one that distinguishes the question’s focus from its presupposition. Similarly, the key semantic processes involve retrieving the correct meaning of the substantive terms in the questions. Finally, the major pragmatic processes involve making inferences about the intent of the question and identifying which features of the question are intended to be meaningful. All three sets of processes can give rise to misunderstandings or other problems.

10.3.2.1. Syntactic problems There are two major obstacles that may prevent respondents from correctly parsing the grammatical form of the question: the question may map onto multiple underlying representations (that is, it may be syntactically ambiguous); or it may be so complex that it overloads working memory so that the representation of the question’s structure is incomplete or incorrect. In our experience, syntactic ambiguity is relatively rare in surveys, though an attitude question from the General Social Survey illustrates the problem (see Fillmore, 1999, for a more extended discussion of this example): “Given the world situation, the government protects too many documents by classifying them as SECRET and TOP SECRET.” The ambiguity involves the topic of the question — is it about the world situation (which is threatening and forces the government to withhold more documents than it otherwise might) or about the government (which classifies too many documents given a relatively benign world situation)?

A more common problem involves excessive grammatical complexity. Consider these questions on disability that appeared on the Census 2000 long form questionnaire:

Because of a physical, mental, or emotional condition lasting 6 months or more, does this person have any difficulty in doing any of the following activities:

- a. Learning, remembering, or concentrating?
- b. Dressing, bathing, or getting around inside the home?
- c. Going outside the home alone to shop or visit a doctor's office?
- d. Working at a job or business?

Each of these four questions encompasses two clauses, the second of which involves multiple possibilities in a disjunctive list. It wouldn't be surprising if some respondents had difficulty in creating a representation of the question that correctly captured all of its logical requirements. Certain grammatical structures are known to impose particularly high burdens on working memory (and therefore should usually be avoided in surveys). For example, embedded clauses can be hard to deal with ("Was that the cat that chased the rat that frightened Karen?") and they have a high risk of overloading working memory. A problem that is closely related to grammatical complexity is *logical complexity*. Even if the syntax of the Census 2000 disability question could be simplified somehow, a single question covering problems in learning, remembering, and concentrating still requires several judgments from the respondent — Does this person have a difficulty of any of the relevant types? Is the difficulty due to a physical, mental, or emotional condition? Is the condition going to last (or has it already lasted) six months or more? Keeping all these logical requirements in mind and carrying them out correctly may be difficult for some respondents.

10.3.2.2. Semantic problems Another set of potential problems involves the meaning of the questions. Many of these semantic problems involve the meaning of individual words. Surveys sometimes use technical terms or other words that respondents may not know. (A growing percentage of Americans do not speak English as their first language and may have trouble even with relatively common English terms.) The authors of survey questions are often substantive experts on the survey topic and they may overestimate respondents' knowledge of the terms used by experts. For example, in a questionnaire on health care, it is tempting to use terms like "PPO," "HMO," and "fee-for-service," but many respondents may be unfamiliar with these terms. The problem of unfamiliar or technical terminology is all the more serious because evidence indicates that respondents often ignore definitions when written definitions are provided with the question (Conrad, Couper, Tourangeau, & Peytchev, 2006; Tourangeau et al., 2006).

Another set of problems derives from the fact that words can have more than one meaning. At the one extreme, the "same" word (e.g., "table") may be represented in the dictionary by several lexical entries or by a single entry with several related senses (as with the word "table"). Apart from these problems of lexical ambiguity, words

are often vague. Consider the word "children." It is polysemous, with two distinct senses (young people regardless of whether they are one's offspring and one's offspring regardless of their age). In addition, the "young people" sense of the word is not precisely defined; depending on the setting, "children" may be defined as people under 21, under 18, or under 12 years of age. Unfortunately, as Belson discovered, different respondents may apply different age cutoffs in understanding the term "children." Survey researchers often use vague quantitative terms as response categories for questions eliciting likelihood ("certain," "very likely," "somewhat unlikely," "impossible," and so on) or frequency ("always," "often," "rarely") judgments rather than asking respondents to make precise quantitative judgments. Again, there is evidence that respondents systematically differ in how they use vague quantifiers, with highly educated respondents making sharper distinctions among terms like "not too often" and "very often" than respondents with lower levels of education (Schaeffer, 1991).

Even seemingly clear-cut terms can be ambiguous or vague. For example, all three of the key terms in the question "Do you own a car?" can create potential comprehension problems. "You" is ambiguous (it can be singular or plural); "own" may mean own outright or merely hold the title to; and "car" may be construed broadly (as any personal vehicle including pickups, motorcycles, or SUVs) or more narrowly (an automobile). As with "car," almost every concept includes both clear cases and borderline instances and respondents may be uncertain as to where to draw the boundary. For example, a traditional four-door sedan is clearly a car but what about a mini-van? Different respondents may handle borderline cases of a concept in different ways, a problem sometimes referred to as *conceptual variability*. Although studies have shown that interviewers can effectively resolve many of these problems (e.g., Conrad & Schober, 2000; Schober & Conrad, 1997), many survey researchers are reluctant to allow interviewers the level of discretion needed for them to clarify vague conceptual boundaries.

A final class of semantic problems involves questions with faulty presuppositions. As we noted earlier, questions present an incomplete picture of some state of affairs and ask the listener to complete that picture. A partial picture that is inaccurate can create problems for the respondent. For example, the attitude item "Family life often suffers because men concentrate too much on their work" presupposes that men concentrate too much on their work, and some respondents, of course, may disagree with that presupposition. If they give a "disagree" response, it indicates that they disagree with the statement, not with its presupposition. Frequency items may also have faulty presuppositions. Asking respondents how often they have done something (witnessed a crime, say) presupposes that they have done it at least once and leads them to report higher frequencies than first asking them whether they have ever witnessed a crime and then asking them how often (Knäuper, 1998). Even though there is a response ("Never") that denies the presupposition of the question, the presupposition still seems to affect the answers.

10.3.2.3. Pragmatic problems Misunderstanding of the questions can also take place as respondents attempt to infer the intent of the question. Their inferences are

not always correct. Couper, Tourangeau, and Kenyon (2004) distinguish between the task and stylistic elements in web questionnaires. The task elements are those that are essential to answering the questions; they include the wording of the question and any accompanying instructions. The stylistic elements are conventions that give the survey its characteristic “look-and-feel” and include the background color of the screen, any logos, images in the header or elsewhere, and so on. Couper and his colleagues argue that respondents may mistakenly infer that elements intended to be stylistic are task elements instead. In their study, respondents’ answers to questions were significantly affected by the pictures that were presented with the questions. As Schwarz and his colleagues have argued, respondents often appear to assume that *every* feature of the questionnaire is intended to be meaningful, including the numerical labels assigned to the points of a rating scale (Schwarz, Knäuper, Hippler, Noelle-Neumann, & Clark, 1991a), the range of the response options (Schwarz & Hippler, 1987), and the shape used to represent the response categories (Schwarz, Grayson, & Knäuper, 1998). For example, Schwarz, Grayson, and Knäuper asked students about which category best represented their academic performance. The question displayed the answer categories in one of three formats — as a ladder, a pyramid, or an onion. The three shapes implied different intended distributions, and, as Schwarz and his colleagues predicted, the responses tended to conform to these implied distributions. Similarly, Tourangeau and his colleagues have demonstrated that the shading and spacing of response options affect answers to questions in a web survey (Tourangeau, Couper, & Conrad, 2004, 2007). Respondents follow Grice’s conversational maxims and they assume that the researchers do too. Thus, they see minor or incidental features of the questions as meaningful, affecting how they answer the questions.

10.3.2.4. Summary As this section makes clear, there are many things that can lead respondents to misunderstand survey questions:

1. The syntax may be ambiguous (mapping onto more than one underlying representation).
2. The syntax (or the underlying logic of the question) may be too complicated, leading to working memory overload.
3. The wording may be too difficult for some respondents or involve technical terms the respondents don’t know.
4. The words in the question may have multiple meanings.
5. The words may be vague, respondents may differ in how they interpret them (that is, there may be conceptual variability across respondents), or the question may involve borderline instances of otherwise clear concepts.
6. The question may be based on a faulty presupposition, one that is not true for the respondent.
7. Respondents may make unintended inferences about the meaning of incidental features of the question.

One of the reasons that cognitive interviewing and other methods of pretesting survey questions are so widespread is that survey researchers are aware of the need to detect and repair these potential sources of misinterpretation of the questions.

10.3.3. Forgetting and Telescoping

10.3.3.1. Forms of forgetting Many survey questions ask about things that must be retrieved from memory, such as dates ("When did you move into this house?"), frequencies of different types of behavior ("How many times did you go to the movies in the last three months?"), or unique events ("Who did you vote for in the last election?"). We have already noted two reasons why people may be unable to recall the information — they may not have encoded the information in the first place (which seemed to account for the poor reporting about children's immunizations) or they may have encoded it, but in terms so different from those used in the question that they are unable to retrieve the information. But there are at least three additional reasons for memory error:

1. Information gets harder to retrieve over time (perhaps because of the accumulation of similar experiences).
2. Retrieval fails to come up with the information (even though the information has been stored in long-term memory).
3. Attempts to infer what happened or to fill in missing details ("reconstruct" the memory) introduce errors instead.

Perhaps the clearest finding in all of cognitive psychology is that it is harder to remember something the more time that has elapsed since it occurred. Despite more than 100 years of research on the impact of time on memory, the exact form of the "retention" curve has not been determined; both the amount of information about an event retained over time and the exact shape of the retention curve depend on the type of event. Distinctive events, important events, and events that last longer are more likely to be remembered over time than common, routine, or brief events.

Apart from characteristics of the events, characteristics of the retrieval attempt can increase or decrease the chances that respondents will remember the information they are seeking. First of all, retrieval takes time. Giving respondents more time to answer questions produces more accurate reports, particularly for behavioral questions. But time on task is not all there is to it. As we noted earlier, memories for autobiographical events are organized hierarchically in large time periods and extended event sequences, such as vacation or bouts of illness (Barsalou, 1988). Giving respondents cues (typically in the form of a brief phrase) to remind them about the time period or event sequence is more effective than trying to get them to retrieve information about a specific event.

Other types of retrieval cue can be helpful as well. Examples of the target category can be an important aid to recall, though they are not a panacea. Giving respondents a list of magazines that they might have read improves reports about which ones they

have read; similarly, a list of types of organization helps respondents remember the organizations they belong to. While examples may help reduce omissions, they have the effect also of being direct cues for memory and result in greater reports for the types of items on the list. On the other hand, if an important type of activity or event is omitted from a list, the absence of a cue for that activity may result in underreporting of the omitted type. The cuing effect of question wording can scarcely be overestimated.

A final form of forgetting involves retrieving incorrect information. Memories often incorporate erroneous information, some of it drawn from things we heard or inferred about the event after it happened. Retrieval is an active process and the cues we generate often represent guesses about what might have happened; it can be difficult to distinguish these guesses from what really happened, especially when the event happened long ago or was poorly encoded in the first place (Johnson, Hashtroudi, & Lindsay, 1993).

Researchers have examined several methods for improving retrieval processes. We have already noted increasing time on task and providing examples or other retrieval cues. Another strategy is the use of a life events calendar. A life events calendar (or events history calendar) asks respondents about one or more life domains, such as their residential or employment history, during a specific period. Respondents might be asked to give the address of each residence where they lived over the period; or they might be asked to note the name of each employer and other facts about each job they held. This information is recorded on a calendar so that the dates and duration for each employment or residential episode is captured. Life events calendars provide multiple retrieval cues for retrieving other information from the time period covered by the calendar and the evidence indicates that they improve recall (Means & Loftus, 1991). Belli (1998) has argued that, relative to other approaches for providing retrieval cues, the life events calendar approach most closely reflects (and takes advantage of) the structure of autobiographical memory. He and his colleagues have shown it is effective even in a telephone interview when respondents can't see the information recorded on the calendar (Belli, Shay, & Stafford, 2001a).

10.3.3.2. Telescoping When thinking about memory problems, we mostly think about forgetting, or the failure to retrieve information. Sometimes, however, we retrieve incorrect information that results in reporting too many events. The best-known example is the phenomenon called *telescoping* (e.g., Neter & Waksberg, 1964). Telescoping errors occur in response to questions about behavior within a bounded time period such as: "How many times have you been to the doctor in the past six months?" Telescoping refers to reporting an event that took place before the period covered by the question (the *reference period* for the question). Memory for the dates of events is particularly poor; dates are one event characteristic that we don't generally encode. In analyzing data from an experimental survey asking about home repairs, Neter and Waksberg estimated that about 30 to 40 percent of the home repair jobs reported in their survey were reported in error due to telescoping. They inferred this from the difference between bounded and unbounded interviews in the number of

reported repair jobs. In a bounded interview, the interviewer reminds respondents of events they already reported in a previous interview, preventing them from telescoping them into the current reference period. Unbounded interviews don't include this review of previously reported events and yield a much high level of reporting.

The most widely accepted explanation for the phenomenon of telescoping involves the loss of information about event dates as the events get older. It is easy to date something that happened yesterday, but much harder to date something that happened several months ago. Memory for the time of events (like memory for other characteristics of events) becomes worse the further back in time the event happened, even though there is no systematic bias in the perceived time of occurrence. Telescoping seems to reflect a combination of this increasing uncertainty about when the events occurred and the imposition of a boundary on the earliest reports. The effect of this bounding is to truncate reports of events that are remembered as having occurred before the beginning of the reference period. Since there is greater uncertainty about when the events occurred the further back in time they actually took place, a larger number of events will be telescoped forward and incorrectly remembered as falling into the reference period than will be telescoped backward out of the reference period. (See Huttenlocher, Hedges, & Bradburn, 1990, and Rubin & Baddeley, 1989, for more detailed discussions of telescoping.)

Apart from bounding, the use of landmark events — major public events that respondents are likely to recall well — to help define the beginning of the reference period reduces telescoping errors. Personal landmark events are helpful as well; for example, respondents might be asked to recall some personally important event that took place near the beginning of the reference period. Even if respondents can't remember the exact date of a landmark event (Wright, Gaskell, & O'Muircheartaigh, 1997), they can often judge whether an event occurred before or after it and this information about relative order can reduce telescoping errors.

10.3.3.3. Seam effects Another manifestation of forgetting, called the "seam effect," occurs in longitudinal or panel surveys (Jabine, King, & Petroni, 1990; Kalton & Miller, 1991). Many longitudinal surveys ask respondents about events that have taken place since the previous interview. For example, if interviews are conducted every four months, the questions typically ask about the previous month (say, April), two months ago (March), three months ago (February), and four months ago (January). The next interview covers the next four months, the period beginning with May and ending with August. The consistent finding is that month-to-month changes are larger for the adjacent months covered by two different interviews (April and May) than between adjacent months (March and April) covered by the same interview; that is, there is a greater difference between the reports of what happened in two consecutive months when the reports come from two different interviews (at the "seam" between the periods covered by the two interviews) than when they come from reports from the same interview.

Seam effects appear to be a function both of memory loss and a particular strategy for reconstructing what happened when memory is poor. The strategy has been called *constant wave responding*. If the questions place a heavy demand on memory,

respondents have the best recall for the most recent period; they reconstruct earlier months by extrapolating backward from the most recent month. For example, they assume their income for prior months is the same as their current income. This strategy reduces the number of changes reported within a given interview. The pileup of changes at the seam is created by differences between the most recent portion of the period covered by the prior interview (for example, April, which is the best remembered month in that interview) and the earliest portion of the next interview (May, which is the mostly likely to be reconstructed). Rips, Conrad, and Fricker (2003) provide a more detailed discussion of seam effects. Seam effects can be reduced by separating questions that ask about the same topic for several months in a row and by providing material (cues, examples, reminders of past behaviors, etc.) that reduces the difficulty of the memory task.

10.3.4. Problems Involving Estimation Processes

The estimation strategies in surveys (at least for behavioral frequency questions) seem to fall into four major categories — retrieval of an exact tally, recall-and-count (and similar bottom-up strategies), rate-based estimation (and related top-down strategies), and impression-based processes (cf. Table 10.1). Because the use of exact tallies seems to be rare, we focus here on the latter three strategies.

The recall-and-count strategy entails recalling specific episodes and is prone to both forms of memory error we distinguished earlier. In fact, the discussion in

Table 10.1: Four classes of judgment/estimation strategies and examples for two types of question.

Strategy	Behavioral frequency questions	Attitude questions
Existing answer	Retrieve <i>exact tally</i> of the quantity	Retrieve <i>existing evaluation</i> of the issue or person
Top-down	<i>Rate-based estimation</i> : Retrieve typical rate and extrapolate over reference period	<i>Value-based judgment</i> : Infer position from more general value or principle
Bottom-up	<i>Recall-and-count</i> (also known as episodic enumeration): Retrieve individual episodes and count them up	<i>Recall-and-integrate</i> : Retrieve specific considerations about the issue or person, evaluate each one, and average those evaluations
Impression	Translate vague <i>impression</i> to number or range	Give <i>impression-based evaluation</i>

Neter and Waksberg (1964) seems to assume respondents answer the question using a recall-and-count strategy. The count may be too low because of retrieval failure (leading to omitted events) or too high (because of telescoping). At least in some cases (including Neter and Waksberg's study), the balance may be in favor of overcounts. Several variables may affect the overall direction of the errors, including variables that improve retrieval (e.g., shorter reference periods, richer retrieval cues, life event calendars, and so on) and those that reduce telescoping (e.g., bounded interviewing or the use of memorable landmark events to bound the beginning of the reference period).

Rate-based estimation can also lead to overestimates or underestimates. Errors are likely to arise when there are departures from the usual pattern from which the rate is derived. We are supposed to visit the dentist twice a year but many of us actually end up going less often than that because of delays or missed appointments; or we may end up going more often than twice because of emergency visits. Respondents may try to compensate for these exceptions to the usual rule, but the initial estimate based on the typical rate may serve as an anchor that respondents fail to adjust adequately. Although some have argued that the use of rate-based estimation characteristically leads to underestimation, we think this conclusion is premature. It is also possible that the rate-based strategy can be implemented imperfectly, leading to errors. For example, multiplying a weekly rate by four to get a monthly total is inherently biased (since months include more than four weeks). Still, rate-based estimates can be quite accurate for respondents who do have a regular rate and use that rate correctly in estimating the total number of events during the reference period (Menon, Raghubir, & Schwarz, 1995).

Impression-based estimates are likely to be quite inexact and associated with the largest reporting errors. Because they are typically unbounded at the high end, they may be gross overestimates. In addition, impression-based estimates are likely to be reported as round values.

10.3.5. Attitude Judgments and Context Effects

Although many nonsurvey judgments are contextual (even simple perceptual judgments regarding the size or brightness of a stimulus can depend on what other stimuli are nearby), it is worrisome to many survey researchers that the answers to a survey question may be strongly affected by the context of the questions. In principle, the context of a survey question could be defined rather broadly (for example, to include interviewer characteristics or the description of the survey's topic or sponsor), but most methodological work on context effects has focused on the questions that immediately precede the question of interest. Many of these methodological studies demonstrate that the order of the questions can alter the answers that respondents give.

Schwarz and Bless (1992) have presented a useful theory of some of the main processes that give rise to context effects in surveys. They argue that in forming their

judgments of an attitude object (say, a political figure), respondents first form a representation of the object, then a representation of a standard, and finally make their judgments by comparing the two. Prior questions can activate concepts that are subsequently included in the representation of the target attitude object; this type of effect is called an assimilation effect. The impact of any one question on the representation of the target attitude object will decrease as the number of other context questions increases. For example, answering a question about marital happiness had a pronounced effect on answers to a subsequent question about general life satisfaction when respondents' marriages were the only specific life domain that the earlier questions asked about. When the earlier questions also asked respondents about their leisure time and their jobs, the effect of the marital happiness item on answers to the question on general life satisfaction was significantly reduced (Schwarz et al., 1991b).

Information activated by earlier questions can be included or excluded from the representation of the target attitude objects that come later in the questionnaire. When information that is ordinarily included in the representation of the target is excluded from it this will lead to a subtraction-based contrast effect. In this case, if the information excluded is positive, the judgment will become more negative; if the excluded information is negative, the judgment will become more positive. The size of the effect depends on the amount and extremity of the temporarily accessible information that is excluded from the representation of the target. Information rendered accessible by prior questions can play an additional role in formulating judgments. In addition to being included in or excluded from the representation of the target, the information may be used in constructing the standard to which the target object is compared; this can produce comparison-based contrast effects. Which of these processes is responsible for a contrast effect determines whether the contrast effect is limited to the target object or generalizes across other objects from the same category. If the contrast effect is based on simple subtraction, the effect is limited to that particular target. If the contrast effect is based on a comparison, the effects are apt to appear in each judgment to which the standard of comparison is relevant.

An example of a contrast effect based on information from previous questions is provided in a study by Schwarz, Münkler, and Hippler (1990). Respondents were asked to rate a number of beverages according to how "typically German" they were. When this question was preceded by a question about the frequency with which Germans drink beer or vodka, contrast effects appeared in the typicality ratings. Respondents who first estimated their consumption of beer (an item Germans consume with high frequency) rated wine, milk, and coffee as less typically German than respondents who first estimated their consumption of vodka; the contrast effect extended across the three target drinks. This effect, however, did not appear when the preceding question was about the caloric content of beer or vodka, presumably because the information activated by this question was not relevant to a judgment about typicality.

Useful as Schwarz and Bless's inclusion-exclusion model is, it does not explain every type of context effect. For example, judgments may also be affected by norms activated by a series of related questions. The juxtaposition of the questions brings to

mind a norm that might not otherwise be applied in making the judgment. This effect was noted in early studies by Cantril and his colleagues (Cantril, 1944). Before the United States entered World War II, they asked respondents in the United States a pair of parallel questions about their willingness to allow Americans to enlist in the British and German armies. More respondents were willing to allow Americans to enlist in the German army when that question followed the question about enlisting in the British army than when order of the two questions was reversed. Similar effects were subsequently observed in studies by Hyman and Sheatsley (1950), Schuman and Ludwig (1983), and Schuman and Presser (1981). In each case, the questions involve policies toward groups of differing popularity; the juxtaposition of the questions invokes the norms of evenhandedness, which mainly affects how respondents treat the less popular group. When the more popular group comes first (e.g., the British), respondents treat it one way (American should be allowed to enlist); then, when the respondents are asked about the less popular group (the Germans), they realize that evenhandedness requires the same treatment to be extended to that group as well.

Feelings evoked by the interview may also affect judgments. For example, thinking about a happy or sad event may influence one's mood. The mood, in turn, may influence subsequent judgments, even though the targets of the judgments are not related to whatever elicited the mood. A number of studies have shown that respondents' mood may influence such things as reported satisfaction with consumer goods (Isen, Shalcker, Clark, & Karp, 1978), feelings toward other people, or estimates of risk (Johnson & Tversky, 1983). The effect of mood on judgments is stronger when the respondents are not aware that their mood is influencing their responses. When the effect of mood is brought to their attention, respondents seem to be able to discount their feelings (Schwarz & Clore, 1983).

10.3.6. *Formatting the Answer*

Survey questions take three common formats. Some questions ask for numerical estimates in an open format:

1. Now, thinking about your physical health, which includes physical illness and injury, for how many days during the past 30 was your physical health not good?

A second type of question asks for a rating on a scale:

2. Would you say that in general your health is excellent, very good, good, fair, or poor?

A final type of survey question asks respondents to choose one answer from among a list of possible answers. In contrast to scales, the options in this final type of question have no inherent order:

3. Are you married, divorced, widowed, separated, never married, or a member of an unmarried couple?

A major difficulty with open-ended questions that ask for numerical answers is that respondents often find it hard to translate their estimate or judgment into an exact number. Consider the task faced by respondents asked the first of the three items above. They are unlikely to have kept a running tally of the number of days when their health seemed less than good. They might have a vague sense that they had a good or bad month or that they were sick for a few days, but the question asks for a specific number of days. Answers to quantitative open-ended questions like this one are often expressed as round values. Respondents are likely to give round answers to questions on such diverse topics as how many sexual partners they have had, how many days ago they completed an interview, or even how old they are (see Tourangeau, Rips, & Rasinski, 2000, Chapter 9, for a review). It is as if the respondents simplify the mapping task by treating the number system as though it consisted of discrete categories (Schaeffer & Bradburn, 1989). The use of round values introduces not only inexactness into the answers but can introduce bias as well. For example, a study by Huttenlocher et al. (1990) found some evidence that respondents tended to round in a biased way, preferring to round down more often than they round up.

Rating scales also have their problems. Many judgments are implicitly based on comparisons to the norms for the relevant category (so that a "tall" five-year-old is likely to be much shorter than a "short" ten-year-old). As a result, the respondents' use of the rating scale will depend on the objects the respondents are asked to rate, with the extreme stimuli serving to define the meaning of the ends of the scale. When respondents are asked to judge a single object (e.g., the performance of the current president), they are likely to define the scale by thinking about other objects of the same type, and which ones they think about are likely to affect both their judgment and how they map it onto the rating scale. Respondents' use of rating scales can also be affected by positivity bias (the tendency to give out mostly favorable ratings), response contraction (the tendency to avoid extreme ratings), and the numerical labels or hues assigned to scale points (see Section 10.3.2 above).

Questions that offer two or more categorical options (such as question 3 above about marital status) are prone to response order effects, in which the order of the response options affects their relative popularity. Sometimes respondents seem to favor the early options (*primacy* effects), sometimes those that were presented last (*recency*). Response order effects appear to reflect limitations on respondents' willingness or ability to process all the options. Working memory is limited and respondents may not be motivated enough to consider every option carefully. Instead, they tend to give more attention to the options they consider first than to those they consider later. At the extreme, respondents may select an answer without even considering some of the options. For instance, a respondent might choose the "never married" option in question 3 above without waiting to hear the final choice ("member of an unmarried couple"), which may be the best answer. This account (which posits progressively less attention to successive options) would suggest that most response order effects would be in the direction of primacy, but recency effects are common as well. The explanation is that when the questions are administered aurally, respondents are likely to begin by considering the last option they heard

rather than the first one presented. What matters is the order in which respondents process the options rather than the order in which the questionnaire or interviewer presents them.

According to Krosnick's (1999) satisficing model response order effects are just one manifestation of satisficing, in which respondents take various shortcuts to minimize the cognitive burden of responding. Response order effects are the product of weak satisficing (in which respondents still retrieve material from long-term memory in formulating their answers). Acquiescence bias (the tendency to agree with statements) is another form that weak satisficing can take. Strong satisficers skip the retrieval step completely, giving "don't know" or "no opinion" responses, repeating the same answer for every item in series of questions with the same response format, or simply giving random guesses.

10.3.7. Response Editing

National surveys sometimes include questions about some very sensitive subjects. The federal government sponsors an annual survey, the National Survey on Drug Use and Health (NSDUH), to monitor the prevalence of illicit drug use. NSDUH is one of the largest surveys done each year in the United States. It asks respondents about their use of a wide range of illicit drugs. Other federally sponsored surveys ask national samples of women whether they have ever had an abortion and national samples of adults whether they voted in the most recent election. Unfortunately, there is a lot of evidence that respondents don't always answer such questions truthfully. Studies that compare self-reports about drug use with results from urinalyses find that some 30–70 percent of those who test positive for cocaine or opiates deny having used the drugs recently (Harrison, 1995). Similarly, studies comparing survey reports about voting with voting records reveal that more than 20 percent of the nonvoters say that they voted in the survey (Belli, Traugott, & Beckmann, 2001b). The reporting errors on these topics are very large and produce biases that are probably the largest source of error in the national estimates derived from these surveys.

Several lines of evidence indicate that much of the error in reports about sensitive topics reflects a more or less deliberate "editing" process, in which respondents alter their answers before they report them. First, with many sensitive topics, almost all the reporting errors are in the socially desirable direction. For example, fewer than 2 percent of those who actually voted incorrectly report that they haven't voted, according to the record checks done as part of the American National Election Studies; by contrast, more than 20 percent of the nonvoters falsely reported that they had voted (Belli et al., 2001b). Similarly, studies that use breath or saliva samples to check the accuracy of reports about smoking find that very few teenage nonsmokers claim to have smoked, but almost half of the teen smokers deny it (Bauman & Dent, 1982; Murray, O'Connell, Schmid, & Perry, 1987). If forgetting or some other problem were responsible for the errors, we would expect to see roughly equal rates

of error in both directions. A second line of evidence comes from studies that examine procedures designed to reduce respondents' motivation to misreport, such as self administration of the questions. Self administration and related methods such as the randomized response technique (Horvitz, Shaw, & Simmons, 1967; Bradburn et al., 1979) take the interviewer out of the equation and increase respondents' willingness to provide otherwise embarrassing information about themselves. These methods have a big effect on responses to sensitive questions, but limited impact on answers to non-sensitive questions. Similarly, methods that increase the respondents' motivation to tell the truth, such as the bogus pipeline procedure (Jones & Sigall, 1971; Tourangeau, Smith, & Rasinski, 1997), in which respondents are convinced that some device or biological assay will reveal any untruthful answers they give, have greater impact on responses to sensitive than to nonsensitive questions. Finally, the changes in reporting produced by the bogus pipeline or self-administration are largest among those respondents with something to hide; for example, the bogus pipeline affects the reported drug use of illicit drug users, but not the reports of those who haven't used illicit drugs.

Two motives seem to underlie this editing process. First, respondents appear to be reluctant to make sensitive disclosures to an interviewer, presumably because they don't want to embarrass themselves. This motive is relevant whenever an interviewer is aware of the significance of their answers (as in telephone or face-to-face interviews with direct questions about the sensitive topic). Second, respondents are sometimes concerned that bystanders and other third parties might become aware of their answers. For example, a meta-analysis by Tourangeau and Yan (2007) suggests that the presence of parents during an interview greatly reduces reporting of sensitive information by their children. Aquilino, Wright, and Supple (2000) argue that the presence of other people affects answers to sensitive questions whenever the bystander doesn't already know the information and the respondent has some reason to fear repercussions should the bystander learn of it. Respondents may edit their answers for other reasons as well. For example, they sometimes seem to tailor their answers to avoid giving offense to the interviewer — they tend to report more profeminist attitudes to women interviewers than to men (Kane & Macaulay, 1993) and more favorable attitudes on civil rights to African American interviewers than to white ones (Hatchett & Schuman, 1975–76; Schuman & Converse, 1971).

The two most practical methods for reducing editing of the answers are self-administration of the questions (including both paper self administration and administration of the questions directly to the respondents via computer) and conducting the data collection in a private setting (at least out of the presence of authority figures, such as parents or supervisors).

10.3.8. Navigation Problems

When an interviewer administers the questions, he or she can help the respondent in various ways, conveying instructions or providing clarification when the respondent

30 a. LAST YEAR, 1999, did this person work at a job or business at any time?

☐ Yes

☐ No → Skip to 31

b. How many weeks did this person work in 1999?

30 a. LAST YEAR, 1999, did this person work at a job or business at any time?

☐ Yes

☐ No → Go to 31

b. (If Yes) How many weeks did this person work in 1999?

Figure 10.2: Two methods for conveying branching instructions. The top method was used in the long form questionnaire in Census 2000; the bottom approach was tested in Census 2000 and reduced the number of branching errors.

doesn't seem to understand the question (although strict standardization may sharply limit the ability of interviewers to give unscripted help to respondents). With self-administered paper questionnaires, such as those used in mail surveys, the respondent has to interpret not only the questions but also any accompanying instructions or definitions. In addition, it is up to the respondents to figure out *which* questions they are supposed to answer. This may require them to follow *skip patterns* — instructions like “GO TO 16” or “Skip to Section B.” Respondents don't always follow such instructions. Instead, they make both of the possible types of navigational errors, sometimes answering questions they should have skipped and sometimes skipping questions they should have answered. Redline, Dillman, Dajani, and Scaggs (2003) report that about 20 percent of the respondents in an experimental panel in Census 2000 made navigational errors in completing the Census 2000 long form.

Redline, Dillman and their colleagues (Redline & Dillman, 2002; Redline et al., 2003) have examined a variety of techniques for reducing respondent errors in completing self-administered questionnaires. The most effective method for getting respondents to follow branching instructions seems to be what they call the prevention method, which used a variety of cues (boldface and larger font for the verbal skip instruction, an arrow pointing to the next question, and an instruction at the item that was to be skipped by respondents) to convey the appropriate navigational path (see Figure 10.2, adapted from Redline et al., 2003).

Jenkins and Dillman (1997; see also Redline & Dillman, 2002) offer several recommendations for the design of self-administered questionnaires, including the following:

1. Use visual elements consistently throughout a questionnaire to define the desired path (for example, the question numbers are given in black circles in Figure 10.2).

2. Any departures from these visual conventions should be signaled prominently (see, for example, the bold skip instruction and use of arrows in the bottom panel of Figure 10.2).
3. Place directions where they are to be used and make them as visible as possible.
4. Present information that needs to be used together in one place.

They also recommend using visual contrast to highlight spaces where respondents are to enter their answers, a convention that is followed by the questions in Figure 10.2.

10.4. Conclusion

The CASM model and related approaches that draw on concepts and theories from cognitive psychology have shed considerable light on a wide range of survey measurement problems, including telescoping, seam effects, question wording and order effects, response order effects, and estimation processes for frequency questions. Perhaps the main contributions of the cognitive approach to survey methodology have been to increase awareness of measurement error and to shift attention from analyzing the consequences of errors to understanding their causes (Tourangeau, 2003). The increased sensitivity to measurement error in surveys has in turn led to the widespread adoption of new methods for pretesting questionnaires. The standard battery of pretesting tools now includes focus groups, cognitive interviews, and expert appraisals, as well as conventional field tests (Presser et al., 2004); these methods were used less frequently (or not at all) 30 years ago.

Still, there are many unsolved problems and potential new areas of application for the cognitive approach. For example, several researchers have begun to tackle the question of how to best use cognitive interviewing to detect and repair survey questions (e.g., Beatty, 2004; Conrad & Blair, 2004; Willis, 2005). Cognitive interviewing, a set of techniques loosely based on methods introduced by Ericsson and Simon (1980), has won widespread acceptance from survey researchers, despite the absence of much evidence that it produces reliable and valid data on questionnaire problems. Further work is definitely needed to investigate when the "problems" detected in cognitive interviews are in fact problems and what types of probes or combinations of probing techniques are most likely to produce valid conclusions. And, although reasonably detailed models of the response process are available for some types of survey questions (e.g., behavioral frequency questions and attitude questions), for most other types little or no empirical work has been done.

The field may also be ready for a switch to other problems and other methodological approaches for dealing with them. Survey response rates are falling throughout much of the developed world and almost all methodologists are concerned about this issue. Worries about nonresponse have generated a conference, a volume (Groves, Dillman, Eltinge, & Little, 2002), and a special issue of

Public Opinion Quarterly all exclusively focused on that topic (see also Dixon & Tucker, this volume). It is not yet clear what insights the cognitive approach may offer on the psychology of *nonresponse*. In addition, methodologists have begun to look to other fields for help with survey methods issues. For example, a recent volume presented papers that examine the survey interview from the perspective of conversational analysis (Maynard, Houtkoop-Steenstra, Schaeffer, & van der Zouwen, 2002), spurred in part by various challenges to the practices of strict standardization (e.g., Schober & Conrad, 1997; Suchman & Jordan, 1990). Conversational analysis represents an alternative approach to understanding the processes that take place in survey interviews, one that is likely to yield new insights into some long-standing methodological issues.

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